

Portal (customer) & transfer approvals

1. Feature Overview

Customer self-service **portal** plus staff **transfer approvals**:

- Portal users (`CUSTOMER` role, `users.customer_id` set) see own accounts, transactions, beneficiaries, and can transfer.
- Transfers may execute immediately or create a **pending** `transfer_request` (threshold or `OTHER_BANK`).
- Staff with accounts write access resolve pending items; history on Approvals screen and Audit.

Status: Implemented (portal + beneficiaries + approvals). Other-bank approve does **not** yet move external ledger (recorded approval only).

2. Business Purpose

Let customers bank online; escalate risky transfers to staff.

3. User Workflow

Customer

1. Login → routed to `/portal/accounts` .
2. View account / transactions.
3. Manage beneficiaries; submit transfer (beneficiary or quick account #).
4. Outcome: `EXECUTED` or `PENDING_APPROVAL` .

Staff

1. `/app/transfer-approvals` — pending queue + **my** resolution history.
2. Approve / reject (reject reason optional).
3. Full history also under Audit → Approval history.

4. Execution Flow

Portal transfer

```
PortalController.transfer
    ↓
PortalTransferService.transfer
    ↓
needsApproval? → TransferRequestEntity PENDING
                : → AccountService.transfer (SAME_BANK)
```

Staff approve (SAME_BANK)

```
TransferApprovalService.approve
    ↓
AccountService.transfer
    ↓
status EXECUTED + audit TRANSFER_APPROVED
```

5. Database Tables

beneficiary

Customer payees: SAME_BANK (linked account) or OTHER_BANK (IFSC/bank name).

transfer_request

Pending/resolved portal transfers: amount, bank type, destination snapshot, requested_by, resolved_by, reasons, timestamps.

customers.transfer_approval_threshold

Optional amount gate for same-bank portal transfers.

users.customer_id

Links portal login to customer.

6. REST APIs

Portal — /portal (ACCESS_PORTAL_ACCOUNTS)

Method	Path	Notes
GET GET GET POST	/portal/accounts /portal/accounts/{id} /portal/accounts/{id}/transactions /portal/accounts/{id}/transfer	Own accounts Own account Paged ledger Body: beneficiaryId or toAccountNumber + amount Active list Create Soft deactivate

Transfer approvals — /transfer-approvals (ACCESS_ACCOUNTS_WRITE)

Method	Path	Notes
GET GET	/transfer-approvals /transfer- approvals/my- history /transfer- approvals/{id}/ approve /transfer- approvals/{id}/ reject	Pending Current user's resolutions

7–13. Code map

PortalController, PortalTransferService, PortalAccountService, PortalCustomerContext,
BeneficiaryService, TransferApprovalService, entities/repos under beneficiary / transfer.

Frontend: `features/portal/*`, `features` approvals screen, `core/services/portal.ts`.

14. Validation Rules

- Portal can only touch own accounts.
- OTHER_BANK always needs approval.
- Same-bank needs approval when amount $\geq$ customer threshold (if set).
- Cannot add own account as beneficiary; quick transfer to own account blocked (own-account move still planned).

15. Security Rules

Portal authority separate from staff; JWT roles drive Angular `portalGuard` vs staff layout.

16–18. Exception / Logging / Audit

Portal transfer request / execute and beneficiary changes write `audit_event` (PORTAL / TRANSFER categories).

19. Testing Strategy

- Low-amount same-bank → EXECUTED
- High-amount / other-bank → PENDING → staff approve
- Reject → REJECTED, no ledger move

20. Future Extension Guide

Own-account internal move; external settlement saga; idempotency keys — see TECH_LEARNING_PLAN.

Future Modification Guide

Requirement: Portal own-account move

Item	Detail
Files	<code>PortalTransferService</code> (lift own-account block); UI transfer mode Still uses <code>AccountService.transfer</code> ; audit PORTAL_TRANSFER